

Samuel’s tensor-algebra formula for the flag structure constants is the Monk-labelled Bruhat chain enumerator

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Abstract

In 2018 Matt Samuel posted (MathOverflow 313951) a formula presenting the structure constants $c_{u,v}^w$ of $H^*(\mathrm{Fl}_n)$ as the expansion coefficients of a “skew” chain sum $T_{w/u}$ in a tensor algebra built on transpositions modulo $(a, b) + (b, c) = (a, c)$. Richard Stanley commented that it “could be the same as Theorem 2 of [math/0202090](#)” (Lenart–Sottile, *Skew Schubert polynomials*); Samuel replied that the two are “very closely related” and that “there may be a transformation from one formula to the other.” Nobody built it.

This note does three things. First it settles the ambiguity in the phrase “strong order”: Samuel’s order is *Bruhat* order graded by Coxeter length, and the absolute-order reading is false, with an explicit witness in S_3 . Second it identifies the formula completely: writing $T_{w/u}$ in the simple-root basis, the coefficient of $\alpha_{p_1} \otimes \cdots \otimes \alpha_{p_m}$ is the number of saturated Bruhat chains from u to w admitting the *Monk labels* $\mathbf{p}$ (Lemma 5). Samuel’s tensor algebra is therefore dual-basis bookkeeping for Monk’s rule, his identity is a coefficientwise family of labelled-chain counts, and it follows from iterated Monk plus commutativity of H^* (Theorem 7). Third, the relation $(a, b) + (b, c) = (a, c)$ is not a convenience but is *forced*, already in degree 1, by Monk’s rule (Theorem 10); the smallest witness is $u = 132$, $w = 231$ in S_3 .

The relation to Lenart–Sottile is then reduced to a single checkable property of their chain weight: *is it multilinear in the root of each cover?* If yes, their Theorem 2 is a specialisation of Samuel’s identity along an explicit map; if no, no formal transformation of that shape exists in either direction (Proposition 13). Deciding this requires the printed definition of “increasing labelled chain,” which I do not hold; the gap is stated precisely in §8, together with a scoped negative result: of 110 candidate local chain rules, the only ones reproducing $\mathfrak{S}_v$ at $n = 3, 4$ are the classical Billey–Jockusch–Stanley compatible sequences, which use *adjacent* covers only.

1 The two sides, with locators and extraction levels

Side A (Samuel). MathOverflow question 313951, “*Does anyone know of this manifestation of the Littlewood–Richardson coefficients for the complete flag variety?*”, Matt Samuel, 2018-10-27, 0 answers, score 8, tags [ag.algebraic-geometry](#), [co.combinatorics](#), [schubert-calculus](#). Samuel attributes the formula to the coproduct for divided-difference operators in his dissertation, and reports that after finding it he found “a proof that was extremely trivial.” Comment thread: Sam Hopkins (“looks like the Monk’s/Chevalley rule way of defining Schubert polynomials... summing over paths from the identity to w in the strong order”); Richard Stanley (“Your result could be the same as Theorem 2 of [arxiv.org/pdf/math/0202090](#)”); Samuel

(“It is very closely related, and I need to look at this a bit harder. There may be a transformation from one formula to the other”).

Extraction level: I hold this at chunk-read — a near-verbatim record of the question body and the comment thread in my own reading log of 2026-10-04 c4, taken from the MathOverflow API. I have not re-fetched the page in this session (prove sessions do not browse).

Side B (Lenart–Sottile). `math/0202090`, Cristian Lenart and Frank Sottile, *Skew Schubert polynomials*, 2002-02-10. Abstract (verified by me at arxiv.org/abs/math/0202090 during browse c4, from the arXiv meta tags): skew Schubert polynomials “expand in the basis of Schubert polynomials with nonnegative integer coefficients that are precisely the structure constants of the cohomology of the complex flag variety with respect to its basis of Schubert classes,” extending Bergeron–Sottile’s construction “in terms of certain increasing labeled chains in Bruhat order of the symmetric group,” and rederived combinatorially “relating it to the construction of Schubert polynomials in terms of rc-graphs.”

*Extraction level: **abstract** only. I do not hold the paper, and I cannot browse in a prove session. Every statement below about Side B is therefore either (i) quoted from that abstract, (ii) derived by me from scratch, or (iii) explicitly marked as a gap. Nothing below cites “Theorem 2” for its content.*

2 Setup

Let S_∞ be the group of finitely supported permutations of $\mathbb{Z}_{>0}$, ℓ the Coxeter length (number of inversions), $s_p = t_{p,p+1}$ the simple reflections, and t_{ab} ($a < b$) the transposition exchanging a and b . We use *right* multiplication throughout: $u \triangleleft w$ is a Bruhat cover iff $w = u t_{ab}$ for some $a < b$ with $\ell(w) = \ell(u) + 1$. Permutations act on positions, so $u t_{ab}$ is u with the entries in positions a and b exchanged.

Schubert polynomials $\mathfrak{S}_w \in \mathbb{Z}[x_1, x_2, \dots]$ are normalised by $\mathfrak{S}_{w_0} = x_1^{n-1} x_2^{n-2} \cdots x_{n-1}$ for $w_0 \in S_n$ and $\mathfrak{S}_{ws_j} = \partial_j \mathfrak{S}_w$ when $\ell(ws_j) = \ell(w) - 1$, where $\partial_j f = (f - s_j f)/(x_j - x_{j+1})$. The structure constants are defined by

$$\mathfrak{S}_u \mathfrak{S}_v = \sum_w c_{u,v}^w \mathfrak{S}_w, \quad c_{u,v}^w \in \mathbb{Z}_{\geq 0},$$

and for $u, v, w \in S_n$ these are the structure constants of $H^*(\mathrm{Fl}_n)$ in the Schubert basis. They are symmetric, $c_{u,v}^w = c_{v,u}^w$, and vanish unless $\ell(u) + \ell(v) = \ell(w)$.

Definition 1 (Samuel’s algebra). Let $W = \bigoplus_{a < b} \mathbb{Q} \tau_{ab}$ be the free $\mathbb{Q}$ -vector space on the transpositions of S_∞ , let

$$R = \mathrm{span}\{ \tau_{ab} + \tau_{bc} - \tau_{ac} : a < b < c \} \subseteq W,$$

and put $V = W/R$, writing t_{ab} for the image of τ_{ab} . Let $F = T(V) = \bigoplus_{m \geq 0} V^{\otimes m}$ be the tensor algebra.

Lemma 2 (“the root system thinly disguised”). *V is free on $e_i := t_{i,i+1}$, $i \geq 1$, and $t_{ab} = e_a + e_{a+1} + \cdots + e_{b-1}$. Hence V is the span of the simple roots of type A_∞ under $e_i \mapsto \alpha_i$, with $t_{ab} \mapsto \alpha_{ab} := \alpha_a + \cdots + \alpha_{b-1} = \varepsilon_a - \varepsilon_b$. Moreover*

$$R = \mathrm{span}\left\{ \tau_{ab} - \sum_{p=a}^{b-1} \tau_{p,p+1} : a < b \right\}.$$

Proof. In V the relation gives $t_{ac} = t_{ab} + t_{bc}$ for $a < b < c$; induction on $b - a$ yields $t_{ab} = \sum_{p=a}^{b-1} e_p$, so the e_i generate V . Conversely let L be free on symbols α_i and define $\pi : W \rightarrow L$ by $\pi(\tau_{ab}) = \alpha_a + \dots + \alpha_{b-1}$. For $a < b < c$, $\pi(\tau_{ab} + \tau_{bc} - \tau_{ac}) = 0$ by telescoping, so $R \subseteq \ker \pi$ and π factors through V , sending $e_i \mapsto \alpha_i$. Thus the e_i are linearly independent, and $V \xrightarrow{\sim} L$. For the last claim: each $\tau_{ab} - \sum_{p=a}^{b-1} \tau_{p,p+1}$ lies in $\ker \pi$, hence (being a sum of the generators of R by the same induction) in R ; conversely $\tau_{ab} + \tau_{bc} - \tau_{ac} = (\tau_{ab} - \sum_a^{b-1}) + (\tau_{bc} - \sum_b^{c-1}) - (\tau_{ac} - \sum_a^{c-1})$. The two spans coincide. $\square$

Definition 3 (the chain sums). For $u \leq w$ in Bruhat order put $m = \ell(w) - \ell(u)$ and

$$T_{w/u} := \sum_{\substack{u t^{(1)} t^{(2)} \dots t^{(m)} = w \\ \ell(u t^{(1)} \dots t^{(i)}) = \ell(u) + i \quad \forall i}} t^{(1)} \otimes t^{(2)} \otimes \dots \otimes t^{(m)} \in V^{\otimes m},$$

the sum over saturated chains in the Bruhat interval $[u, w]$, each cover labelled by the transposition effecting it on the right. Set $T_v := T_{v/e}$, $T_{u/u} = 1$, and $T_{w/u} = 0$ if $u \not\leq w$.

Note that if $u, w \in S_n$ then every x with $u \leq x \leq w$ lies in S_n (subword property: x is a subword of a reduced word for w , which uses only $s_1, \dots, s_{n-1}$), so the chain sums for S_n stay inside S_n and only $\alpha_1, \dots, \alpha_{n-1}$ occur.

3 Which order? Settling “strong”

The phrase “strong order” is ambiguous: in Björner–Brenti it is a synonym for Bruhat order (as opposed to the *weak* order), while “the order generated by all reflections” with its own grading is the *absolute* order, graded by reflection length $\ell_R(w) = n - \#\{\text{cycles of } w\}$. Samuel’s modulus is imposed on transpositions, which invites the absolute reading. It is wrong, for a reason visible before any computation.

Proposition 4. *In Definition 3 the grading ℓ must be Coxeter length, i.e. the chains are saturated chains in Bruhat order. With reflection length in place of ℓ the identity $T_{w/u} = \sum_v c_{u,v}^w T_v$ is false.*

Proof. Degrees. The identity equates an element of $V^{\otimes m}$ with $m = \ell(w) - \ell(u)$ on the left to a combination of T_v with $\ell(v) = m$ on the right, and $c_{u,v}^w \neq 0$ forces $\ell(u) + \ell(v) = \ell(w)$ with ℓ the Coxeter length, since that is the cohomological grading of $H^*(\text{Fl}_n)$. The two gradings are genuinely different: $t_{13} = 321 \in S_3$ has $\ell = 3$ but $\ell_R = 1$.

A witness. Take $u = 321$ and $w = 231$ in S_3 . Then $\ell_R(u) = 1$, $\ell_R(w) = 2$, and $u t_{12} = 231 = w$ is an absolute-order cover, so the absolute-order chain sum is $T_{w/u}^{\text{abs}} = t_{12} = \alpha_1 \neq 0$. But $\ell(u) = 3 > 2 = \ell(w)$, so $c_{u,v}^w = 0$ for every v and the right-hand side is 0. $\square$

So *half of the question Stanley raised is not at issue*: both sides are Bruhat-order chain formulas. What remained was the labelling, and that is what the next section identifies.

4 The structural identification: Samuel's tensors are Monk labels

Lemma 5 (dual basis). *For $p \geq 1$ let $\phi_p \in V^*$ be the coordinate functional dual to α_p , so $\phi_p(\alpha_i) = \delta_{pi}$. Then*

$$\phi_p(t_{ab}) = \begin{cases} 1, & a \leq p < b, \\ 0, & \text{otherwise,} \end{cases}$$

and consequently, for $\mathbf{p} = (p_1, \dots, p_m)$,

$$[\alpha_{p_1} \otimes \dots \otimes \alpha_{p_m}] T_{w/u} = (\phi_{p_1} \otimes \dots \otimes \phi_{p_m})(T_{w/u}) = N_{w/u}(\mathbf{p}),$$

where $N_{w/u}(\mathbf{p})$ is the number of saturated Bruhat chains $u = u_0 \leq u_1 \leq \dots \leq u_m = w$, with $u_i = u_{i-1} t_{a_i b_i}$, satisfying

$$a_i \leq p_i < b_i \quad (1 \leq i \leq m). \quad (\text{Monk labels})$$

In particular every coefficient of $T_{w/u}$ in the simple-root basis is a nonnegative integer, and $T_{w/u}$ carries exactly the information of the Monk-labelled chain counts $N_{w/u}(\cdot)$.

Proof. By Lemma 2, $t_{ab} = \sum_{q=a}^{b-1} \alpha_q$, whence the value of ϕ_p . Expanding a single chain monomial,

$$t_{a_1 b_1} \otimes \dots \otimes t_{a_m b_m} = \sum_{\substack{\mathbf{p} \\ a_i \leq p_i < b_i}} \alpha_{p_1} \otimes \dots \otimes \alpha_{p_m},$$

each basis tensor occurring with coefficient exactly 1. Summing over chains gives the claim. $\square$

This is the whole mystery of the tensor algebra. A “term” of Samuel's $T_{w/u}$, read in the simple-root basis, is a saturated Bruhat chain *together with a Monk label for each cover* — an integer p_i in the half-open interval $[a_i, b_i)$ cut out by the i -th reflection. The relation $(a, b) + (b, c) = (a, c)$ is precisely the statement that the label set of t_{ac} is the disjoint union of those of t_{ab} and t_{bc} .

Lemma 6 (Monk's rule; classical). *For every $p \geq 1$ and $u \in S_\infty$,*

$$\mathfrak{S}_{s_p} \cdot \mathfrak{S}_u = \sum_{\substack{a \leq p < b \\ \ell(u t_{ab}) = \ell(u) + 1}} \mathfrak{S}_{u t_{ab}}, \quad \text{equivalently} \quad c_{u, s_p}^w = \begin{cases} 1, & u \leq w = u t_{ab}, \ a \leq p < b, \\ 0, & \text{else.} \end{cases}$$

Monk (1959); the G/B form is Chevalley's formula at the fundamental weight ω_p , since $\langle \omega_p, \alpha_{ab}^\vee \rangle = [a \leq p < b]$. See Macdonald, *Notes on Schubert polynomials*, (4.15'). Independently re-verified in this session against the divided-difference Schubert instrument: 16/16, 288/288, 7320/7320 instances at $n = 3, 4, 5$ (§7).

Theorem 7 (Samuel, MO 313951). *For all $u, w \in S_\infty$ with $m = \ell(w) - \ell(u) \geq 0$,*

$$T_{w/u} = \sum_{v: \ell(v)=m} c_{u,v}^w T_v \quad \text{in } V^{\otimes m}.$$

Proof. Write $Y_p := \mathfrak{S}_{s_p} = x_1 + \cdots + x_p$ and $Y_{\mathbf{p}} := Y_{p_1} Y_{p_2} \cdots Y_{p_m}$. By Lemma 5 the asserted identity is equivalent to the family of scalar identities

$$N_{w/u}(\mathbf{p}) = \sum_{\ell(v)=m} c_{u,v}^w N_v(\mathbf{p}), \quad \mathbf{p} \in \mathbb{Z}_{>0}^m. \quad (*)$$

Step 1 (iterated Monk). For all u and all $\mathbf{p}$,

$$Y_{\mathbf{p}} \cdot \mathfrak{S}_u = \sum_w N_{w/u}(\mathbf{p}) \mathfrak{S}_w. \quad (**)$$

Induct on m . For $m = 0$ both sides are $\mathfrak{S}_u$ and $N_{u/u}() = 1$, $N_{w/u}() = 0$ for $w \neq u$. For $m \geq 1$ write $\mathbf{p} = (p_1, \mathbf{p}')$. Since the polynomial ring is commutative,

$$Y_{\mathbf{p}} \mathfrak{S}_u = Y_{\mathbf{p}'} (Y_{p_1} \mathfrak{S}_u) \stackrel{\text{Lem. 6}}{=} \sum_{\substack{u < u' = u t_{ab} \\ a \leq p_1 < b}} Y_{\mathbf{p}'} \mathfrak{S}_{u'} \stackrel{\text{ind.}}{=} \sum_w \left(\sum_{\substack{u < u' = u t_{ab} \\ a \leq p_1 < b}} N_{w/u'}(\mathbf{p}') \right) \mathfrak{S}_w.$$

A Monk-labelled chain from u to w with label word $\mathbf{p}$ is exactly a first cover $u < u' = u t_{ab}$ with $a \leq p_1 < b$ followed by a Monk-labelled chain from u' to w with label word $\mathbf{p}'$, so the inner sum is $N_{w/u}(\mathbf{p})$. This is $(**)$.

Step 2 (the same product, expanded in the other order). Put $u = e$ in $(**)$ and use $\mathfrak{S}_e = 1$:

$$Y_{\mathbf{p}} = \sum_{\ell(v)=m} N_v(\mathbf{p}) \mathfrak{S}_v,$$

the sum being over $\ell(v) = m$ because $Y_{\mathbf{p}}$ is homogeneous of degree m . Multiplying by $\mathfrak{S}_u$ and expanding,

$$Y_{\mathbf{p}} \mathfrak{S}_u = \sum_{\ell(v)=m} N_v(\mathbf{p}) \mathfrak{S}_v \mathfrak{S}_u = \sum_w \left(\sum_{\ell(v)=m} c_{v,u}^w N_v(\mathbf{p}) \right) \mathfrak{S}_w = \sum_w \left(\sum_{\ell(v)=m} c_{u,v}^w N_v(\mathbf{p}) \right) \mathfrak{S}_w,$$

using $c_{v,u}^w = c_{u,v}^w$. Comparing with $(**)$ and using that $\{\mathfrak{S}_w\}$ is a basis gives $(*)$. $\square$

Remark 8 (what the proof uses, and what the tensor algebra contributes). Exactly two inputs: Monk's rule, and the commutativity and associativity of $H^*(\text{Fl})$ — the fact that $Y_{\mathbf{p}} \mathfrak{S}_u$ may be computed either by letting the Y 's act on $\mathfrak{S}_u$ one at a time, or by first expanding $Y_{\mathbf{p}}$ in the Schubert basis at $u = e$. The tensor algebra F contributes nothing beyond Lemma 5: it is a device for holding all $(n-1)^m$ Monk-label statistics in one object. This is, I believe, the “extremely trivial” proof Samuel refers to.

Lemma 9 (independence, and uniqueness of the coefficients). *Fix m and n . The matrix $(N_v(\mathbf{p}))$, rows indexed by $v \in S_n$ with $\ell(v) = m$ and columns by $\mathbf{p} \in \{1, \dots, n-1\}^m$, has full row rank. Equivalently $\{T_v : \ell(v) = m\}$ is linearly independent in $V^{\otimes m}$, so the coefficients in Theorem 7 are unique and the theorem may be read as an alternative definition of $c_{u,v}^w$.*

Proof. By Step 2 of the previous proof, $Y_{\mathbf{p}} = \sum_v N_v(\mathbf{p}) \mathfrak{S}_v$, so the row space of the matrix is the space of Schubert-basis coordinate vectors of the products $Y_{\mathbf{p}}$, i.e. of the degree- m part of the subring generated by $Y_1, \dots, Y_{n-1}$. By Borel, $H^*(\text{Fl}_n; \mathbb{Q}) = \mathbb{Q}[x_1, \dots, x_n]/(e_1, \dots, e_n)$, and $x_p = Y_p - Y_{p-1}$ with $x_n = -Y_{n-1}$ (as $e_1 = 0$), so the Y_p generate the whole ring. Hence the row space is all of H^{2m} , of dimension $\#\{v \in S_n : \ell(v) = m\}$, which is the number of rows. $\square$

(Measured: the rank equals 1, 4, 9, 15, 20, 22, 20, 15, 9, 4, 1 for $m = 0, \dots, 10$ at $n = 5$ — the Mahonian numbers, as the lemma predicts; see §7. This is also why the sum in Theorem 7 may be restricted to $v \in S_n$ when $u, w \in S_n$: the coefficients being unique, and an identity holding with only S_n terms, no $v \notin S_n$ can contribute.)

5 The relation is forced

Samuel presents the modulus $(a, b) + (b, c) = (a, c)$ as a definition. It is not a choice.

Theorem 10 (inevitability of R). *Let U be a $\mathbb{Q}$ -vector space and $\pi : W \rightarrow U$ a linear map; write $\bar{t}_{ab} = \pi(\tau_{ab})$ and let $T_{w/u}^U \in U^{\otimes m}$ be the chain sums of Definition 3 formed with $\bar{t}$ in place of t . If*

$$T_{w/u}^U = \sum_{\ell(v)=1} c_{u,v}^w T_v^U \quad \text{for every cover } u \lessdot w,$$

then $R \subseteq \ker \pi$, i.e. π factors through Samuel's V . Thus V is the largest quotient of W in which Theorem 7 can hold, and degree 1 alone forces it.

Proof. Fix $a < b$. There is a u with $u \lessdot u t_{ab}$: take u with $u(a) = 1$, $u(b) = 2$, and the remaining values $3, 4, \dots$ assigned to the remaining positions in increasing order. Then $u(a) < u(b)$, and for $a < c < b$ we have $u(c) \geq 3 > u(b)$, so no value lies strictly between; hence $u t_{ab}$ covers u . Put $w = u t_{ab}$.

In degree 1 there is exactly one chain from u to w , with label t_{ab} , so the left-hand side is $\bar{t}_{ab}$. On the right, $\ell(v) = 1$ means $v = s_p$, and $T_{s_p}^U = \bar{t}_{p,p+1}$; by Lemma 6, $c_{u,s_p}^w = [a \leq p < b]$. Hence the hypothesis reads

$$\bar{t}_{ab} = \sum_{p=a}^{b-1} \bar{t}_{p,p+1}, \quad \text{i.e.} \quad \tau_{ab} - \sum_{p=a}^{b-1} \tau_{p,p+1} \in \ker \pi.$$

By Lemma 2 these elements span R . □

The smallest witness is as small as possible. In S_3 , take $u = 132$ and $w = 231$: the unique chain is $u t_{13} = w$, so the left side is t_{13} , while Monk gives $c_{132,s_1}^{231} = c_{132,s_2}^{231} = 1$ and the right side is $t_{12} + t_{23}$. One pair (u, w) , and the defining relation drops out. Without the relation the identity fails at 4 of 13 pairs in S_3 and 110 of 235 in S_4 (§7).

6 The transformation, and what it reduces to

Corollary 11 (the transformation exists, abstractly). *Fix m . By Lemma 9 there is a unique linear map*

$$\Psi_m : \text{span}\{T_v : \ell(v) = m\} \longrightarrow \mathbb{Z}[x_1, x_2, \dots], \quad \Psi_m(T_v) = \mathfrak{S}_v.$$

Theorem 7 says $T_{w/u}$ lies in that span, so $\Psi_m(T_{w/u})$ is defined, and

$$\Psi_m(T_{w/u}) = \sum_{\ell(v)=m} c_{u,v}^w \mathfrak{S}_v.$$

Hence defining the skew Schubert polynomial by $\mathfrak{S}_{w/u} := \Psi_m(T_{w/u})$ yields a statement of exactly the shape the abstract of [math/0202090](#) describes.

Corollary 11 is cheap, and saying only this would be an evasion. It shows a transformation exists, but Ψ_m is defined by inverting a matrix, not by a rule on chains. What a theorem like Lenart–Sottile’s asserts beyond Samuel’s is that $\Psi_m(T_{w/u})$ has a *local, manifestly positive, chain-combinatorial* expression. So the real question is locality, and it sharpens to a single property.

Definition 12. A *chain weight* of length m with values in a commutative ring A is a function ω on m -tuples of transpositions; it defines $\Omega_{w/u} := \sum_{\text{chains}} \omega(t_{a_1 b_1}, \dots, t_{a_m b_m})$. Call ω *root-multilinear* if it is the restriction of an A -valued multilinear form on $V^{\otimes m}$ — equivalently, if in every slot $\omega(\dots, t_{ac}, \dots) = \omega(\dots, t_{ab}, \dots) + \omega(\dots, t_{bc}, \dots)$ for all $a < b < c$.

Proposition 13 (dichotomy). *Let ω be a chain weight.*

- (i) *If ω is root-multilinear, then $\Omega_{w/u} = \sum_v c_{u,v}^w \Omega_v$ holds for all $u \leq w$, and is an immediate specialisation of Theorem 7 along the induced $\phi_\omega \in (V^{\otimes m})^* \otimes A$. Conversely, the family of all root-multilinear specialisations recovers Theorem 7, since such ϕ span $(V^{\otimes m})^*$.*
- (ii) *ω is root-multilinear if and only if there exists a linear ϕ on $V^{\otimes m}$ agreeing with ω termwise, i.e. $\phi(t_{a_1 b_1} \otimes \dots \otimes t_{a_m b_m}) = \omega(t_{a_1 b_1}, \dots, t_{a_m b_m})$ for every tuple. So if ω is not root-multilinear, the Ω -identity is not obtained from Theorem 7 by evaluating a functional chain by chain.*
- (iii) *The weaker demand — that $\Omega_{w/u} = \phi(T_{w/u})$ for some single ϕ and all $u \leq w$, with no termwise condition — is **automatic** and carries no information: if Ω satisfies the identity $\Omega_{w/u} = \sum_v c_{u,v}^w \Omega_v$ at all, then $\phi := \Psi_m$ of Corollary 11 composed with $T_v \mapsto \Omega_v$ works, by Lemma 9.*

Proof of (ii) and (iii). (ii) The tuples $(t_{a_1 b_1}, \dots, t_{a_m b_m})$ map injectively into $V^{\otimes m}$ (a pure tensor determines its factors up to scalars, and each t_{ab} is determined by its support $\{a, \dots, b-1\}$), so a termwise ϕ exists iff ω is consistent with every linear relation among those images — and those relations are generated, slot by slot, by $t_{ac} = t_{ab} + t_{bc}$. That consistency is root-multilinearity. (iii) By Lemma 9 the T_v with $\ell(v) = m$ are independent, so $\phi(T_v) := \Omega_v$ defines a linear map on their span; by Theorem 7 $T_{w/u}$ lies in that span, and $\phi(T_{w/u}) = \sum_v c_{u,v}^w \Omega_v$, which the hypothesis says is $\Omega_{w/u}$. $\square$

Part (iii) is the reason Corollary 11 is cheap: *any* family of “skew” objects satisfying an identity of Lenart–Sottile’s shape is a specialisation of Samuel’s identity in the weak, non-local sense. So the only substantive reading of “there may be a transformation from one formula to the other” is the chain-level one, and by (ii) that reading is exactly root-multilinearity.

Proof of (i). Root-multilinearity means ω factors as $\omega(t_{a_1 b_1}, \dots) = \phi_\omega(t_{a_1 b_1} \otimes \dots \otimes t_{a_m b_m})$ for a (unique) A -valued linear functional ϕ_ω on $V^{\otimes m}$; then $\Omega_{w/u} = \phi_\omega(T_{w/u})$ and applying ϕ_ω to Theorem 7 gives the identity. For the converse, the functionals of product form $\phi^{(1)} \otimes \dots \otimes \phi^{(m)}$ span $(V^*)^{\otimes m} = (V^{\otimes m})^*$, and each is root-multilinear. $\square$

Corollary 14 (Q345 reduced to one check). *Whether Theorem 2 of [math/0202090](#) is a transformation of Samuel’s formula in the only sense that carries information — chain by chain — is decided by a single property of the weight attached to an increasing labelled chain: **is that weight root-multilinear?** Equivalently, in each slot, does the weight of a chain through t_{ac}*

equal the sum of the weights obtained by replacing that cover's label by t_{ab} and by t_{bc} ? If yes, their theorem is Samuel's identity specialised along ϕ_ω , and Stanley's guess is correct in the strongest sense. If no, then by Proposition 13(ii) no chain-level transformation exists in either direction: the two theorems compute the same numbers by mechanisms whose only common point is $c_{u,v}^w$ itself, and the pair is an instance of "same identity, independent derivation" — which is the harder and more interesting answer to Q342.

Remark 15 (my prediction, recorded before I can test it). I expect the answer to be *no*. Two reasons. First, an "increasing" condition couples consecutive covers through their index data (a_i, b_i) , and such conditions are typically not additive under $t_{ac} \rightsquigarrow t_{ab} + t_{bc}$, since t_{ab} and t_{bc} sit differently in any order on reflections than t_{ac} does. Second, and more concretely: at $u = e$ the weight must reproduce $\mathfrak{S}_v$, and the one local rule that does so (Billey–Jockusch–Stanley compatible sequences; see §8) is supported on *adjacent* covers only — and restricting to adjacent covers is manifestly not root-multilinear, as it discards t_{ac} while keeping t_{ab}, t_{bc} . I record this here, in advance and in writing, so that if the printed definition later agrees with it I do not mistake my own prediction's confirmation for independent corroboration.

7 Computational verification

Code and evidence logs: `proofs/code-1004c4-q345/` — `{schub,chains,monk,test_samuel,test2,test_all.r` with the output of each run committed alongside as `*.out`. No SageMath is installed in this container, so everything is pure Python; this turned out to be an advantage, because it forced the ground truth to be built from a different mechanism than the object under test.

Instruments, and their independence.

I1 *Schubert instrument* (ground truth). Schubert polynomials from $\mathfrak{S}_{w_0} = x_1^{n-1} \cdots x_{n-1}$ by divided differences; structure constants $c_{u,v}^w$ extracted from the product $\mathfrak{S}_u \mathfrak{S}_v$ by applying ∂_j along a descent path of w and reading the constant. Knows nothing of chains, Monk, or Chevalley. Sanity: reproduces the textbook S_3 table $\mathfrak{S}_{213} = x_1$, $\mathfrak{S}_{132} = x_1 + x_2$, $\mathfrak{S}_{231} = x_1 x_2$, $\mathfrak{S}_{312} = x_1^2$, $\mathfrak{S}_{321} = x_1^2 x_2$.

I2 *Chain instrument*. Enumerates saturated Bruhat chains by right multiplication, expands $t_{ab} = \alpha_a + \cdots + \alpha_{b-1}$, and returns $T_{w/u}$ as a dictionary on label words.

I3 *Monk transfer matrices*. $M_p[u][v] = 1$ iff $v = u t_{ab}$ is a cover with $a \leq p < b$.

I2 and **I3** are *the same mechanism* — I3 is I2 reorganised as a transfer operator by Lemma 5. Their agreement is an implementation check, not corroboration, and is recorded as such. The only independent check of Side A is against **I1**.

(V1) Monk's rule against I1. Agreement 16/16 ($n = 3$), 288/288 ($n = 4$), 7320/7320 ($n = 5$) instances (p, u, v) with $\ell(v) = \ell(u) + 1$. Zero mismatches. This is hazard H2 discharged: the input to the proof is ground-truthed against the independent instrument before being used.

(V2) Theorem 7 directly, as a tensor identity. Full tensors from I2, coefficients from I1. Bruhat reading: 23/23 pairs (u, w) at $n = 3$, 341/341 at $n = 4$, zero disagreements. Absolute-order reading: 2 disagreements of 25 at $n = 3$, 68 of 385 at $n = 4$; the first is the witness of Proposition 4.

(V3) Negative control (hazard H3). A +1 planted on one structure constant is detected: $n = 3$, perturbing $c_{132,312}^{321}$ gives 1 disagreement of 23; $n = 4$, perturbing $c_{2134,2431}^{3421}$ gives 1 of 341. The comparator can say “different.”

(V4) Exact test for *all* label words of *all* degrees. Because M_p is supported on pairs with $\ell(w) - \ell(u) = 1$, the span A_m of all products $M_{p_1} \cdots M_{p_m}$ is supported on $\ell(w) - \ell(u) = m$; and $(*)$ is linear in that matrix. So it suffices to check it on a basis of A_m , built inductively as $A_m = \text{span}(A_{m-1}M_p)$ and row-reduced over $\mathbb{Q}$. Results (pairs $\times$ $\dim A_m$ checks, all degrees, exact):

n	$\dim A_m, m = 0, 1, 2, \dots$	total checks	disagreements
3	1, 2, 2, 1	35	0
4	1, 3, 5, 6, 5, 3, 1	1115	0
5	1, 4, 9, 15, 20, 22, 20, 15, 9, 4, 1	74199	0

The dimensions are the Mahonian numbers $\#\{v \in S_n : \ell(v) = m\}$, exactly as Lemma 9 predicts. This matters for two reasons. It confirms Lemma 9 empirically at $n \leq 5$; and it shows the test is *not vacuous* — A_m is as large as it can be, so the functionals $X \mapsto X[e, v]$ form a coordinate system on it and no constraint is automatically satisfied.

(V5) Chain instrument against transfer matrices (implementation check only). 50 label words at $n = 3$, 5893 at $n = 4$ (all degrees), 315 138 at $n = 5$ (degrees $m \leq 4$, 7305 pairs). Zero mismatches. This transports the custody of (V4) back to Definition 3 as literally written.

(V6) Theorem 10. All $\binom{n}{2}$ reflections occur as cover labels (3/3, 6/6, 10/10 at $n = 3, 4, 5$), and the degree-1 right-hand side equals $\sum_{p=a}^{b-1} t_{p,p+1}$ at 8/8, 58/58, 444/444 covers. Running the *same* identity in the free space W , with the relation not imposed: 4 of 13 pairs fail at $n = 3$ and 110 of 235 at $n = 4$; smallest witness $u = 132$, $w = 231$, free LHS = τ_{13} , free RHS = $\tau_{12} + \tau_{23}$.

(V7) Reconstruction search (negative, scoped). See §8.

8 Gaps, precisely stated

G1 (the one that matters). I hold math/0202090 at extraction level **abstract**. I therefore do not know the printed definition of “increasing labelled chain,” and cannot evaluate Corollary 14’s single criterion. Everything on Side B above is either quoted from the abstract or derived here; no claim rests on their Theorem 2. *This is a reading task, not a proof task*: one deep read of math/0202090 §1–2, extracting (a) the definition of a labelled chain, (b) the increasing condition, (c) the monomial attached, suffices to decide Q345 by Corollary 14. That is the next action, and it belongs to a browse cycle.

G2 (scoped negative: the naive local rules do not work). To avoid leaving G1 purely as an absence, I searched for the rule by derivation instead of reading, using the constraint that at $u = e$ any such formula must return $\mathfrak{S}_v$. The space: 110 rules of the form (label interval

$\in \{[1, a], [1, b-1], [a, b-1], \{a\}, \{b-1\}\} \times (\text{monotonicity condition on the label word} \in \{\text{free, weak, strict, weakly/strictly decreasing, two Billey–Jockusch–Stanley-style mixed conditions keyed on } a, b \text{ or } b-a\}) \times (\text{all Bruhat covers / adjacent covers only}), \text{ monomial } \prod_i x_{q_i}$. Result: exactly 4 rules reproduce $\mathfrak{S}_v$ for every $v \in S_3$, the same 4 survive every $v \in S_4$, and *all four use adjacent covers only*, where they collapse to one rule — the classical Billey–Jockusch–Stanley compatible-sequence formula. **No rule in the space works with general Bruhat covers.**

Scope, stated honestly. This is a null on an enumerated space of 110 parametrised rules at $n = 3, 4$, not a proof that no local rule exists. It does not refute Side B — Lenart–Sottile’s construction is certainly outside this space (that is the point of their paper) — and it is not evidence about their weight’s root-multilinearity. What it does establish is that the obvious guesses fail, which is why the “transformation” was never one line, and it is the concrete support for Remark 15.

G3. Verification is at $n \leq 5$; Theorem 7, Theorem 10 and Proposition 13 are proved for all n (indeed in S_∞), so this is a check, not a dependency.

G4. Side A is held at chunk-read, from my own browse log, not from a fresh fetch of MO 313951. The mathematical content was independently re-derived and verified here, so an error in my record would have surfaced as a failed test; but the *attribution* — that this is what Samuel wrote — rests on that log.

9 Summary of the answer to Q345

1. **The order is Bruhat order**, graded by Coxeter length. The absolute-order reading is refuted by $u = 321$, $w = 231$ in S_3 . So both sides of Stanley’s comparison are Bruhat-chain formulas, and the question was never about the order.
2. **Samuel’s formula is the Monk-labelled Bruhat chain enumerator.** In the simple-root basis the coefficient of $\alpha_{p_1} \otimes \cdots \otimes \alpha_{p_m}$ in $T_{w/u}$ is the number of saturated chains $u \rightarrow w$ whose i -th reflection $t_{a_i b_i}$ satisfies $a_i \leq p_i < b_i$. The tensor algebra is dual-basis bookkeeping; the modulus $(a, b) + (b, c) = (a, c)$ is the additivity of Monk-label intervals.
3. **It is a theorem, with a three-line proof:** iterated Monk, then commutativity of H^* . Verified exactly at $n = 3, 4, 5$ in all degrees against an independent Schubert-polynomial instrument, with a negative control.
4. **The relation is forced**, already in degree 1, by Monk’s rule; V is the largest quotient of the free space on transpositions in which the identity can hold. Smallest witness: t_{13} versus $t_{12} + t_{23}$.
5. **The transformation question is now one checkable property:** is Lenart–Sottile’s chain weight root-multilinear (Corollary 14)? The weak, non-local sense of “transformation” is automatic and empty (Proposition 13(iii)), so this is the only reading with content. If yes, their theorem is Samuel’s identity specialised along an explicit functional, and the family of all such specialisations is equivalent to his. If no, no chain-level transformation exists in either direction. My recorded prediction is *no* (Remark 15); deciding it needs one deep read, not more computation.

In Pak's terms (Q342): this pair is not yet confirmed-independent nor confirmed-equivalent, but it is no longer vague. The "is there a transformation?" question of 2018 has been replaced by a yes/no property of a single definition, and one of the two sides is now completely understood.